

Which customers get which offer, and what it is worth

Recommendation. Contact 5,000 of 17,059 eligible customers with one offer each, for \$10,000 of incentives. The models expect \$42,649 of value; scored against the simulator's true responses the plan creates **\$64,774, 75% of what a perfectly informed planner could reach.**

The one thing to remember

The customers most likely to travel are not the ones an offer moves. Ranking customers by travel propensity and ranking them by the value an offer truly creates go in opposite directions (rank correlation -0.47). Giving 10% off to the most frequent travellers, with the same budget, creates \$6,278: 7% of the ceiling. Most of that budget discounts trips that would have happened anyway.

The plan in numbers

	Value	What it means
Contacts	5,000	limit 5,000
Incentive spend	\$10,000	budget \$10,000
Expected value	\$42,649	30-day net contribution after incentives, discounted days 31-90 margin, congestion relief
True value (simulation)	\$64,774	75% of the \$85,868 perfect-knowledge ceiling
Rush-hour trips moved onto the 407	+86 per workday	only where the zone has free-flow room at peak
Certificate	proven optimal	one program over 112,563 customer-offer decisions; gap $1.5e-07$ to the bound

Synthetic portfolio simulation inspired by electronic toll-road decisions. Customers, rates, zones, capacities and policies are invented; the weather, holiday and exchange-rate context is real public data. Not affiliated with 407 ETR.

Two randomised tests stand behind the plan

The July offer trial

Every customer was randomised into control or one of 9 offers (2,500 per arm against 1,794 required), with assignment balanced within strata of past travel (worst covariate imbalance 0.033 standardised difference; allocation check $p = 1.00$). Using each customer's June behaviour as a control variable removed 84% of the noise, so the trial could detect an effect of 0.35 trips per customer.

Offer	Net contribution per randomised customer, 30 days	95% interval (Bonferroni)
Fifth trip free	\$1.09	−\$0.62 to \$2.81
Spend & get \$10	\$0.97	−\$0.85 to \$2.79
Weekend 20% off	−\$0.64	−\$2.38 to \$1.10
Off-peak 15% off	−\$0.81	−\$2.56 to \$0.93
10% off every trip	−\$1.34	−\$3.22 to \$0.53
Rush-hour 25% off	−\$2.76	−\$4.57 to −\$0.94
500 bonus points	−\$3.95	−\$5.75 to −\$2.16
Off-peak pass (\$30 cap)	−\$8.01	−\$9.95 to −\$6.07
1,500 bonus points	−\$12.42	−\$14.23 to −\$10.61

Most offers lose money on the average customer: a discount is paid on every trip, including the ones the customer would have taken anyway. That is why the plan targets the customers each offer actually moves rather than sending the best-on-average offer to everyone.

The October policy test

The targeting policy itself was then randomised: 12,500 customers governed by the plan's rules against 12,500 who were not, counting everyone the plan chose not to contact. Result: **\$1.40 net contribution per customer** (95% interval \$0.23 to \$2.57, $p = 0.019$).

Customer models, scored on an untouched later month

Model	Algorithm	AUC	Calibration error	Use
Propensity	xgboost	0.913	0.057	reporting and segments, never targeting
Churn	xgboost	0.910	0.005	inactivity risk in the next 90 days
Attrition	hist gradient boosting	0.740	0.080	risk among recently active customers

Release gate: 8 of 8 checks passed. The gate is evaluated on every run; at 3,000 customers it fails, and the run is not published.

Every limit has an owner and a price

Where the money goes

Offer	Contacts	Spend	Expected value
Fifth trip free	3,000	\$1,927	\$33,839
Spend & get \$10	1,386	\$2,708	\$5,029
Off-peak 15% off	237	\$3,403	\$2,303
10% off every trip	24	\$927	\$815
Weekend 20% off	281	\$801	\$535
Rush-hour 25% off	49	\$221	\$113
Off-peak pass (\$30 cap)	23	\$12	\$14

What relaxing a limit is worth

The optimiser reports, for each limit that binds, how much expected value one more unit would add. That turns a request for more budget or more inventory into a priced trade-off.

Limit	Current	Value of one more unit
Inventory: Fifth trip free	3,000	\$9.43
Incentive budget	10,000	\$0.37
Contact limit	5,000	\$0.25

Rush hour

Peak capacity is a hard limit with a 20% reserve. The busiest peak cells after the campaign: Northwest at 83%; Central at 81%; Northeast at 80%. Rush-hour offers are used only where there is room, and off-peak offers that move commuters out of the peak free capacity for them.

How the plan compares with simpler targeting

Approach	True value	Share of ceiling
Oracle optimum (true effects)	\$85,868	100%
Optimized (MIP)	\$64,774	75%
Uplift ranking (greedy)	\$64,158	75%
Optimized, risk-averse (MIP on LCB)	\$57,727	67%
Propensity: 10% off top travellers	\$6,278	7%
Random targeting	\$1,107	1%
RFM segment playbook	-\$4,226	-5%

Limits that change how to read this

Estimates are conservative in level. The models under-state each customer-offer's value by \$4.49 on average, which is why the expected \$42,649 sits below the simulated true \$64,774. The plan depends on ranking customers correctly, not on the level; the policy test measures the level directly.

The simulation is a test bench, not the business. It shows the method finds value where a simpler rule does not; it cannot show what the real response rates are. The first real campaign should run as a policy test like October's before any full rollout.

The numbers move with the computing environment. On identical data, the same pipeline on Databricks (whose platform fixes older numpy and pandas) fits slightly different models and its plan reaches 65% of the ceiling instead of 75%. Both clear the release gate; read the share as a range.

Congestion relief is a policy choice. It is valued at \$0.50 per net rush-hour trip moved onto the highway; the business should set that number.

Recommended next steps

Step	Owner	Decision it enables
Approve the plan for a randomised pilot	Marketing	Measure real incremental value per customer
Release more free-trip inventory	Marketing	The limit with the highest value per unit
Set the congestion-relief value	Traffic operations	How much rush-hour shifting is worth
Provide dated consent and status history	IT	Point-in-time eligibility for every plan

Where the work runs

Surface	Status
Pipeline and release gate	Runs end to end in 7 minutes; gate passed
Decision app and API	Twelve-page app; decision API with keys and health checks
Power BI	Nine-page report; every measure executed against the Power BI engine
Databricks	Four-task job ran on serverless: Delta, PySpark parity, Unity Catalog, every check passed
AWS SageMaker	Pipeline defined and its stages run locally; not run in AWS